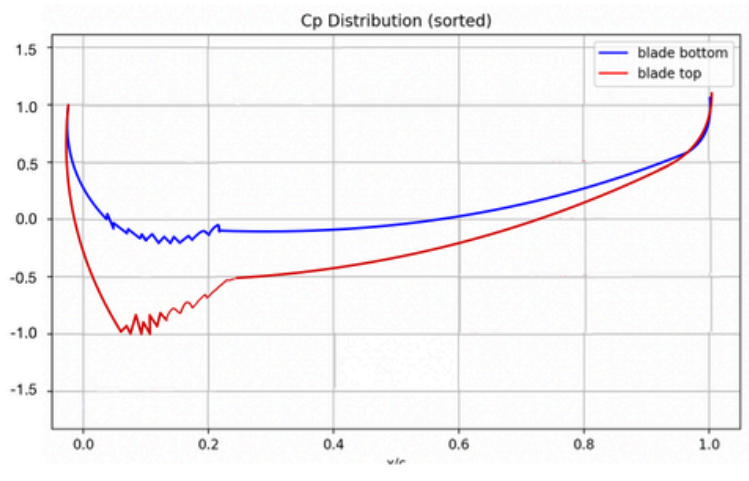
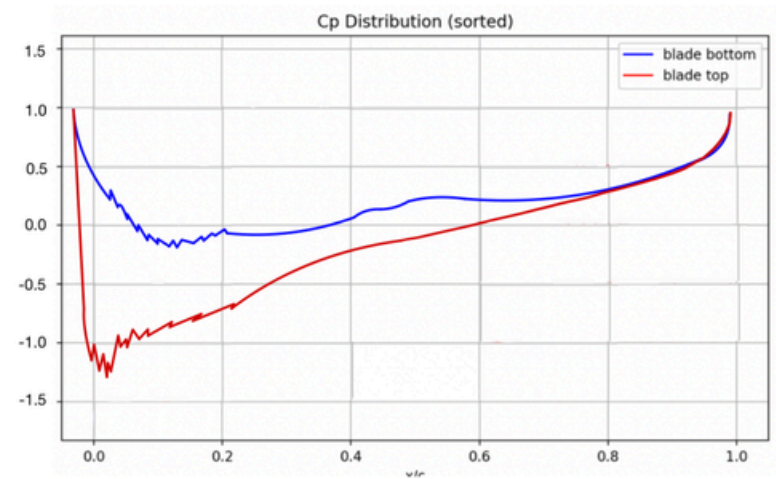


전산유체해석실습 13주차

항공기계공학과
2021010525
원동주



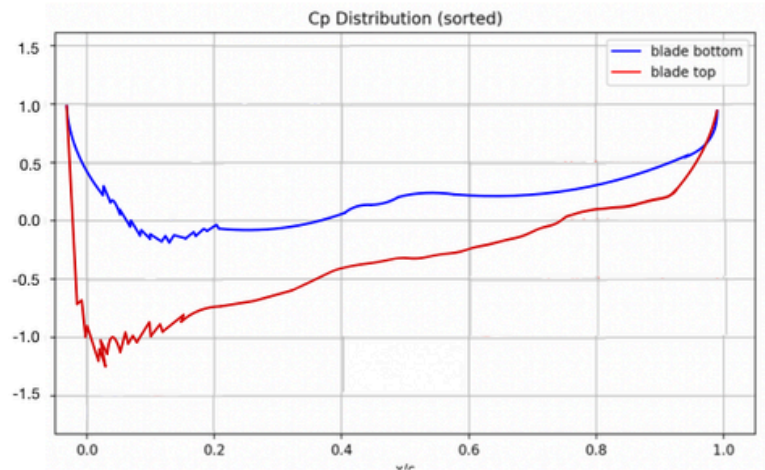
r/R=0.5



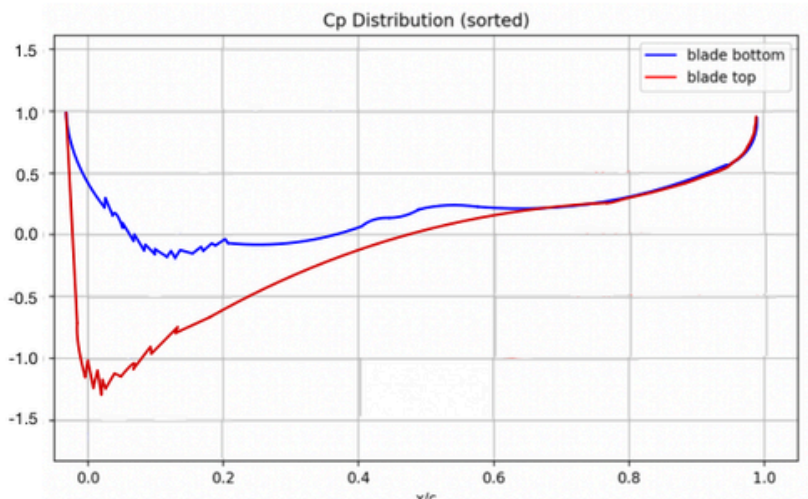
r/R=0.89

```
Users > wondongju > Downloads > try28.py > ...
1  import pandas as pd
2  import matplotlib.pyplot as plt
3
4  # CSV 읽기
5  df = pd.read_csv("slice_11.csv")
6
7  # x/c 계산
8  x = df["Points_0"]
9  df["x_over_c"] = (x - x.min()) / (x.max() - x.min())
10
11 # top/bottom 분리
12 top = df[df["Points_1"] > 0].copy()
13 bottom = df[df["Points_1"] < 0].copy()
14
15 # *** 핵심: x_over_c 기준 정렬 ***
16 top = top.sort_values(by="x_over_c")
17 bottom = bottom.sort_values(by="x_over_c")
18
19 # Cp 값
20 cp_top = top["Pressure_Coefficient"]
21 cp_bottom = bottom["Pressure_Coefficient"]
22
23 # x/c
24 x_top = top["x_over_c"]
25 x_bottom = bottom["x_over_c"]
26
27 # 그래프
28 plt.figure(figsize=(8,5))
29 plt.plot(x_bottom, cp_bottom, 'b-', label="blade bottom")
30 plt.plot(x_top, cp_top, 'r-', label="blade top")
31
32 plt.gca().invert_yaxis()
33
34 plt.xlabel("x/c")
35 plt.ylabel("C_p")
36 plt.title("Cp Distribution (sorted)")
37 plt.grid(True)
38 plt.legend()
39 plt.tight_layout()
40 plt.show()
41 |
```

파라뷰 z normal 값에 각각의 r/R값을 계산해 slice 한 뒤 파이썬 코드를 이용해 그래프 그림.



r/R=0.8



r/R=0.96